

Next-Generation Web Puzzle Platform

Glowtris Unified Master Report

(Complete Edition - EN)

Business Strategy, Technical Architecture & UX/UI Design Specifications

Date: June 14, 2026

To: Executive Board & Research Directors

Classification: Confidential

Chapter 1. Business Feasibility & Marketing Strategy

Executive Summary

This report outlines the final commercialization master plan for 'Glowtris', an ultra-responsive web-based puzzle platform. The project retains the hardcore tactile feel powered by a **1000Hz independent tick engine**, while strategically pivoting towards an **asynchronous retention system inspired by Duolingo** to ensure mass appeal. To guarantee flawless operability, the system adopts a **Vanish Zone Architecture (internal 10x40 logic / visual 10x20 output)**, strictly adhering to modern Tetris standards. Coupled with a **'Dynamic Target Overtake'** UI and robust legal mitigation strategies against global IP risks, Glowtris aims to become a leading puzzle platform characterized by zero server maintenance costs, high user retention, and unparalleled mechanical integrity.

1.1 Strategic Pivot: Asynchronous Competition

By entirely removing real-time netcode PvP matches, the project eliminates infrastructure costs and casual user stress.

- **Zero Server Infrastructure Cost:** A Serverless architecture where only final scores are transmitted to the DB post-client calculation.
- **Dynamic Target Overtake:** Replacing visual noise-inducing 'Ghost Pieces', the UI prominently displays the score of the 'Next Target Ranker'. Surpassing this score triggers a swipe animation demoting the previous target, generating intense micro-motivation and psychological tension.

1.2 SWOT Analysis & Business Feasibility

Strengths	Weaknesses
- Zero server maintenance costs (Maximized profitability) - Flawless control via 1000Hz local engine - High daily retention driven by Duolingo-style UX	- Lack of direct interactive combat (e.g., sending garbage lines) <i>* Mitigation: Compensated by the Dynamic Target Overtake UI</i>
Opportunities	Threats
- Void in the mobile browser-optimized puzzle engine market - High viral potential on short-form platforms (TikTok/Shorts)	- Aggressive IP enforcement by The Tetris Company (TTC) <i>* Mitigation: Robust legal defense strategy applied (See 1.3)</i>

1.3 Legal Mitigation & Go-To-Market Strategy

- **IP Bypass Design:** Complete exclusion of trademarked terms (Tetris, Tetromino, T-Spin). To prevent Trade Dress disputes, the official 7-Bag color scheme is replaced with Glowtris's proprietary deep navy and neon color palette.

- **Data Analytics:** Internalizing UI A/B testing and analyzing the Streak break cycles through the AARRR funnel.
- **Initial Seeding:** Distributing technical dev-logs on hardcore communities (e.g., Reddit) highlighting the '1000Hz sub-frame mobile-compatible engine', and spreading ultra-fast 0-Finesse gameplay footage as 'Oddly Satisfying' content.

Chapter 2. Engine Architecture & Mechanism Control

2.1 Core Gimmick: The 10x40 Vanish Zone Architecture

Design Principle: Maintain the modern 10x20 standard for gameplay feel while masking the internal data structure for system flexibility.

The engine's internal 2D logic array is designed as 10x40 (Width x Height). However, the WebGL client viewport only renders the bottom 10x20 matrix. This architecture prevents immediate Lock-out bugs when blocks spawn atop highly stacked columns and ensures flawless physical collision calculations (Wall Kicks) when blocks rotate above the 20th row ceiling.

2.2 Parity Preservation & 0-Finesse Optimization

- **Parity Preservation:** Supports precise collision algorithms (3-Corner Rule) enabling players to utilize T-Spins to resolve the topological imbalance ($\Delta P = |2|$) caused by the T-Mino.
- **0-Finesse Optimization:** Integrates a lossless Input Buffering system, allowing players to reduce keystrokes to mathematical minimums under the 7-Bag Random Piece Generator system.

2.3 1000Hz Tick Rate & 0ms ARR Discontinuous Transition

Bypassing the browser's 60Hz rendering limitation, the physics engine operates on an independent **1000Hz (1ms) fixed tick**. When ARR (Auto Repeat Rate) is set to 0ms, intermediate loops are bypassed using a **Discontinuous State Transition** formula that instantly teleports the block to the furthest collision coordinate, reducing input latency to zero.

Chapter 3. Behavioral Psychology UX/UI & Retention

3.1 Gestalt Psychology-Based Spatial Separation

- **Law of Similarity (Clickable):** Gamable interactive buttons secure a touch target of at least 48dp, popping out visually via neon highlights and Z-axis drop shadows.
- **Law of Closure (Non-clickable):** Information areas (scores, locked skins) utilize dimmed treatments and translucent blurs to visually signal inaccessible spaces.

3.2 Cross-Platform Layout Adaptability

Mobile Environment (Portrait - 50%)	PC Environment (Landscape - 50%)
- Top: Info (Avatar, Streak, Target Rank) - Center: 10x20 Board Viewport - Bottom 30%: Finger Zone (Controls) - Bottom-most: Tab GNB (Home, Rank, Store)	- Left Panel: Daily Dashboard & Leaderboard - Center Panel: Game Board, Score (Focus) - Right Panel: Match Info, System Settings

3.3 Micro-interactions & IDEAL Framework Validation

All components adhere to the IDEAL framework (Intent, Design, Effectiveness, Action, Learning). Features include Z-axis scale suppression (0.95) on button press, red vignetting flashes during critical situations, and dynamic particle swipe effects upon overtaking a target rank.

3.4 Duolingo-style Gamification Architecture

To transition single-session arcade play into a 'daily habit', a **Streak loop** is utilized. A dynamic avatar system, built via the Factory Pattern, synchronizes with the game's State Machine to exhibit real-time reactions (e.g., cheering, panicking) during gameplay.

Chapter 4. Mobile UI Wireframes (Appendix)

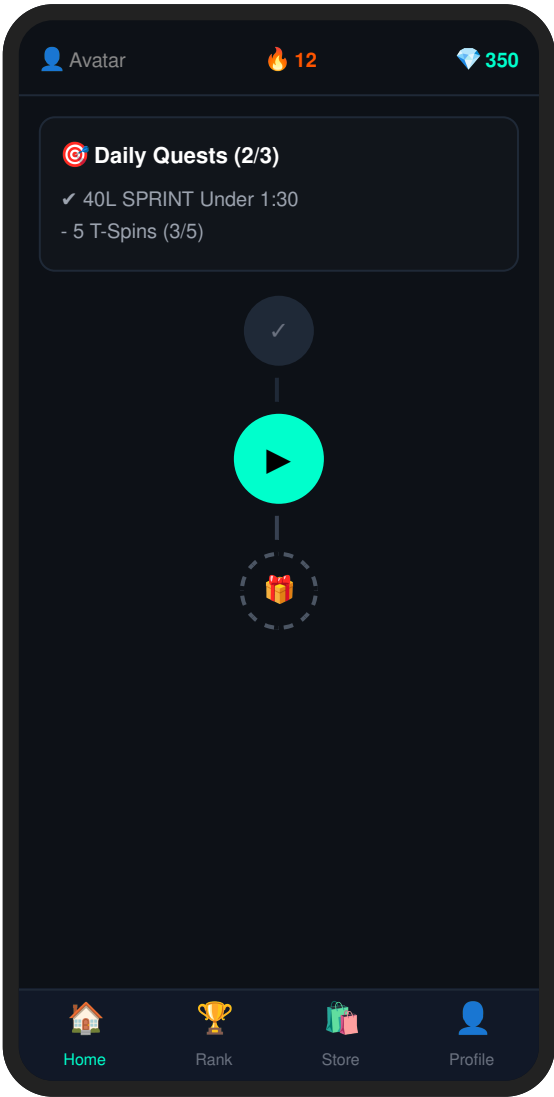


Fig 1.
Home
(Daily

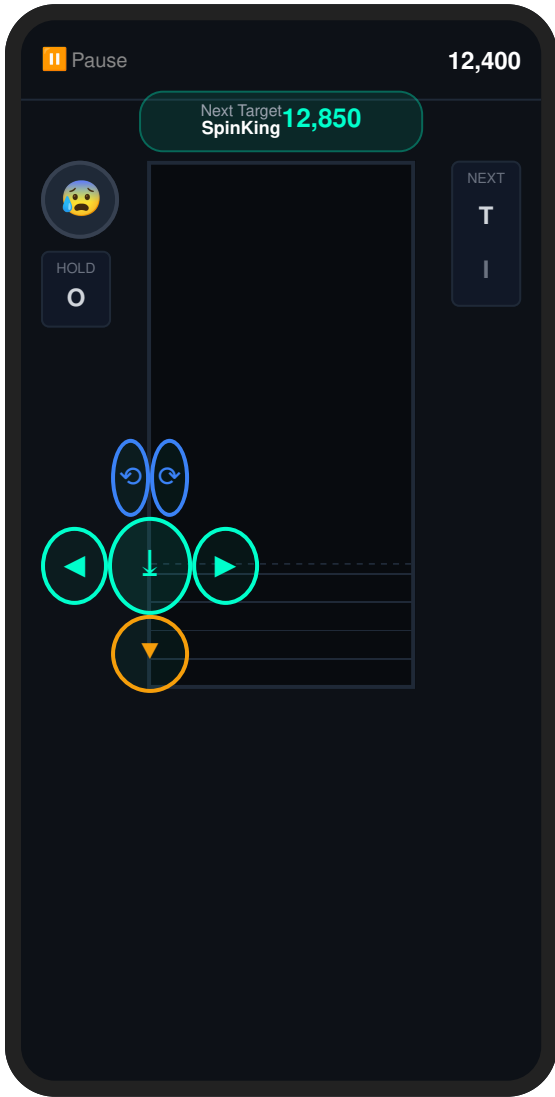
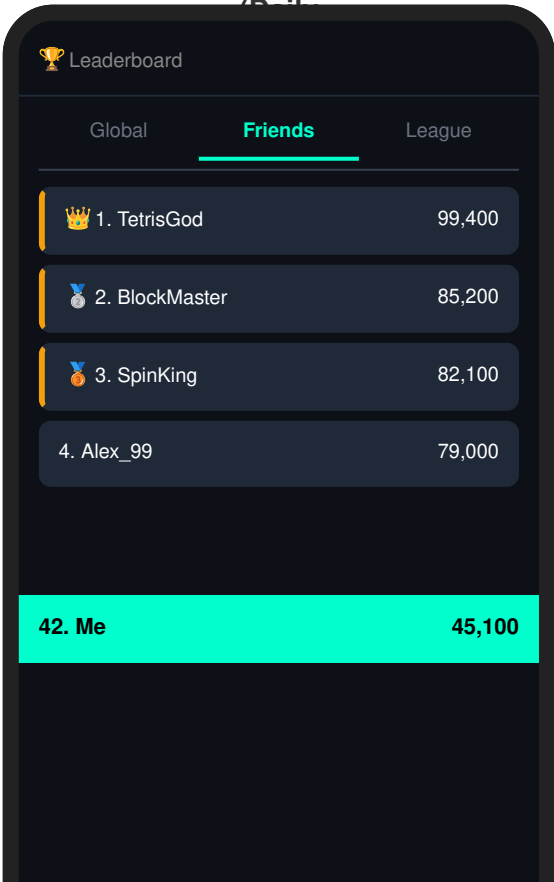


Fig 2. In-
Game
(Target
Overtake
UI)



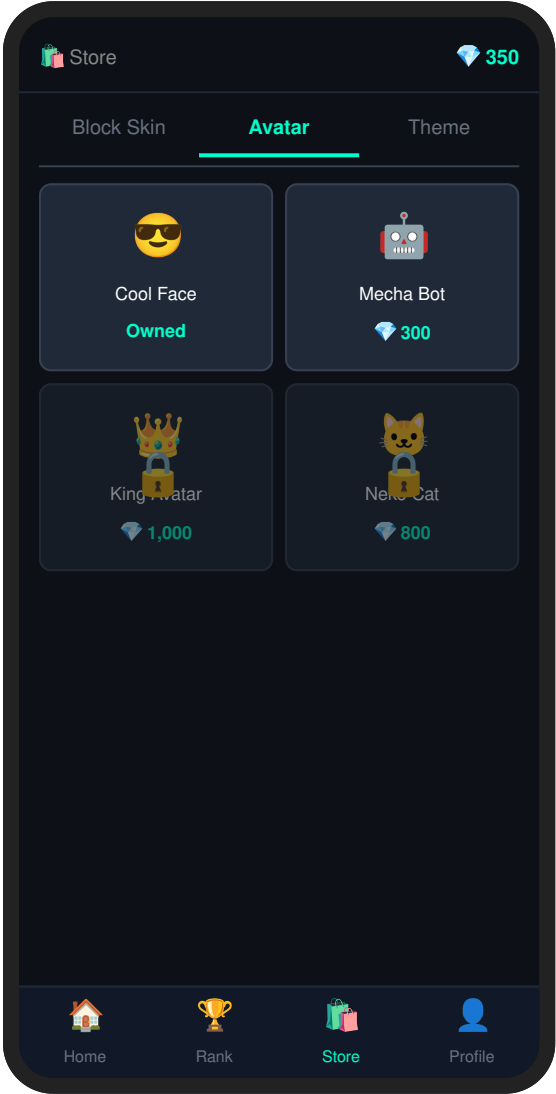
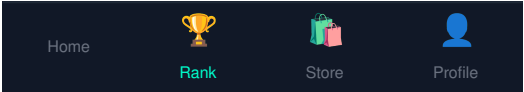


Fig 4.
Modular
Grid
Store



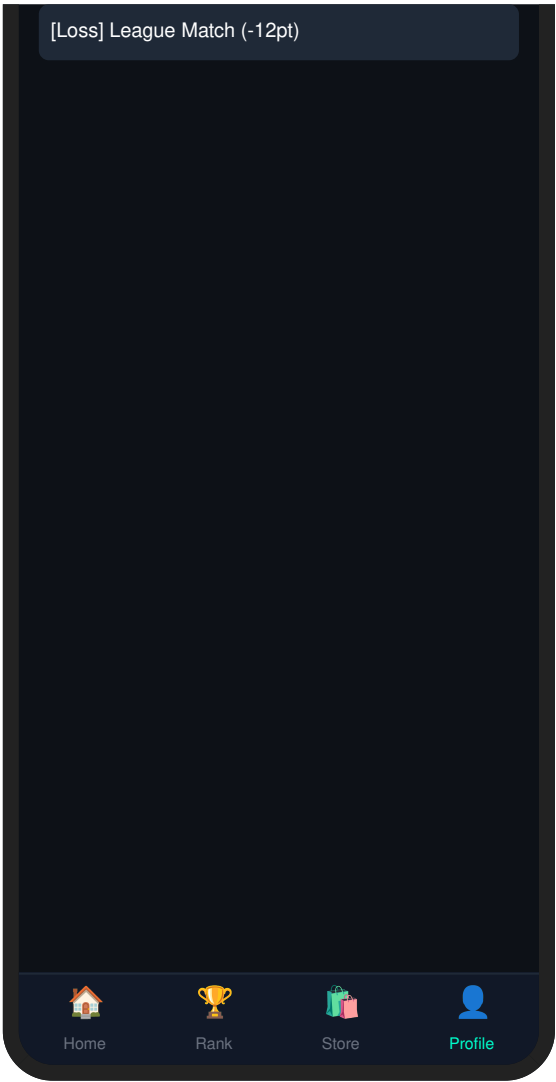


Fig 5.
Profile
&
Data
Stats

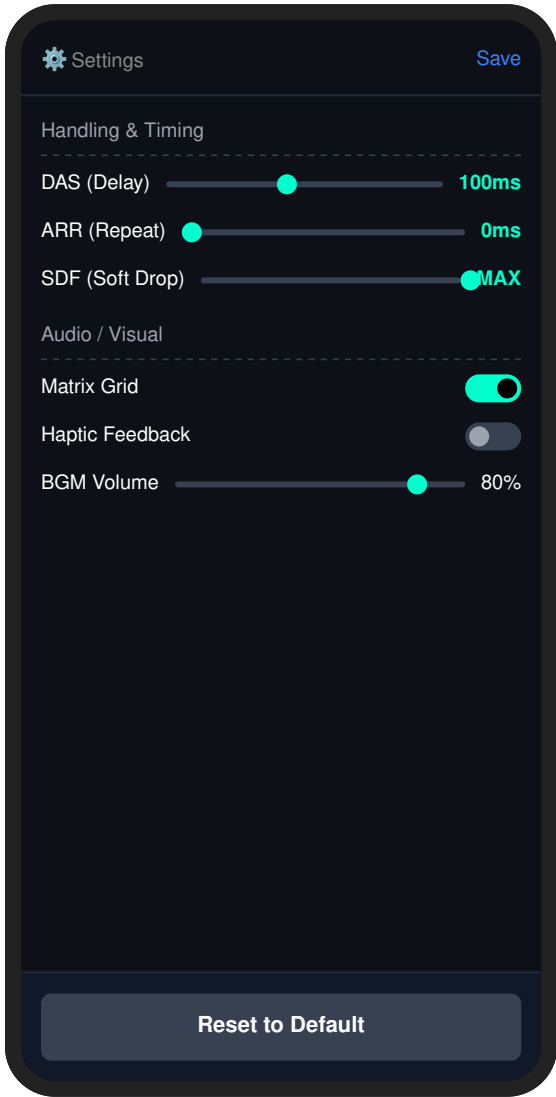


Fig 6. Engine
Customization